



DeepPlaq: Dental plaque indexing based on deep neural networks

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Abstract

Objectives The selection of treatment for dental plaque is closely related to the condition of the plaque on different teeth. This study validated the ability of CNN models in assessing the dental plaque indices.

Materials and methods In 70 (20 male and 50 female) healthy adults (18 to 55 years old), frontal and lateral view intraoral images (210) of plaque disclosing agent stained permanent and deciduous dentitions were obtained. A three-stage method was employed, where the You Look Only Once version 8 (YOLOv8) model was first used to detect the target teeth, followed by the prompt-based Segment Anything Model (SAM) segmentation algorithm to segment teeth. A new single-tooth dataset consisting of 1400 photographs was obtained after applying a two-stage method. Finally, a multi-class classification model DeepPlaq was trained and evaluated on the accuracy of dental plaque indexing based on the Quigley-Hein Index (QHI) scoring system. Classification performance was measured using accuracy, recall, precision, and F1-score.

Results The teeth detector exhibited an accuracy (mean average precision, mAP) of approximately 0.941 ± 0.005 in identifying teeth with plaque disclosing agents. The maximum accuracy attained in the plaque indexing through DeepPlaq was 0.84 (probability that DeepPlaq scored identical to experts), and the smallest average scoring error was less than 0.25 on a 0 to 5 scale for scoring.

Conclusions A three-stage approach demonstrated excellent performance in detecting and segmenting target teeth, and DeepPlaq model also showed strong performance in assessing dental plaque indices.

Clinical relevance Application of artificial intelligence to the evaluation of dental plaque distribution could enhance diagnostic accuracy and treatment efficiency and accuracy.

Keywords Dental plaque · Deep learning · Teeth detection · Teeth segmentation · Classification · Multi-view intraoral images

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Introduction

Dental plaque, comprised of microbial biofilm on tooth surfaces, accumulate predominantly at gingival margins and interproximal areas [1]. Localized plaque accumulation is associated with the initiation and progression of dental caries, gingivitis, and periodontal disease [2–4]. Consequently, periodic assessment of plaque distribution and subsequent interventions are crucial for the prevention and management of dental disease.

Assessments of dental plaque severity, through scoring and indexing [5], are typically performed by trained oral health professionals who possess adequate experience and background knowledge. These experts assign scores ranging from 0 to 5 to indicate the severity of plaque accumulation on each tooth. However, the availability and access to adequately trained and calibrated experts for periodic

monitoring by plaque indexing for each patient's dentition, poses a challenge. Promise lies in the emerging application of artificial intelligence (AI) for dental health diagnoses that are based on deep learning techniques. Clinicians individually inspecting plaque on each tooth is a time-consuming and labor-intensive task. Therefore, deep learning methods were applied to expedite this process. The integration of deep learning with medical issues has accelerated diagnostic and treatment efficiency for clinicians.

Previous studies have explored tasks such as tooth detection, segmentation, and numbering across different types of dental datasets, including two-dimensional intraoral, radiographic images, and three-dimensional cone-beam computed tomography (CBCT) scans [6–8]. For instance, Gerhardt et al. [9] used two subsequent (CNNs) based on the 3D-UNet to detect and segment teeth in CBCT scans. However, this method lacked adequate accommodation for the diverse range of radiological machines and patient variability from complex dental conditions. Xu et al. [10] employed a two-stage deep learning-based approach for tooth detection and numbering that utilized U-Net network to segment regions of interest (ROIs) in radiographic datasets, which was followed by a cascade HTC network to perform further instance segmentation and tooth numbering. But this method failed to effectively address more intricate tooth numbering errors during post-processing. In contrast, Almaliki et al. [11] proposed a denoised encoder-based residual U-Net for accurately segmenting teeth from panoramic radiographic images. Li et al. [6] designed a landmark detector on intraoral images to approximately localize teeth by curvature, followed by fine segmentation using an active contour model aided by key points. However, all of these approaches focused only on the detection and segmentation of anterior teeth, and their performance suffered under certain circumstances, such as missing teeth or larger spaces between adjacent teeth.

The rapid advancement of artificial intelligence (AI) and deep learning algorithms, particularly CNNs, are being integrated into dentistry [12–14]. CNNs are capable of capturing diverse features from images, including edges, colors, and shapes, while filtering out extraneous background details. This attribute proves invaluable for tasks such as tooth detection, segmentation, numbering, and plaque scoring. Hence, the utilization of CNNs is anticipated to minimize variability and human error, thereby enhancing accuracy and expediting diagnostic processes to optimize treatment outcomes [15]. Furthermore, the most recent and advanced model for object detection, YOLOv8, has been adapted for a variety of tasks in medical imaging with remarkable success, including the detection of cells, elbow osteochondritis dissecans, and colorectal lesions, among others [16–18]. Additionally, SAM has been devised to segment objects of

interest within images based on specific prompts provided by users. These prompts may be in the form of a single point, a set of points (including a complete mask), a bounding box, or text [19]. SAM was based upon the vision transformer model, and was trained on a large dataset with 11 million images comprising 1 billion masks [20]. Moreover, SAM is the first foundation model for general image segmentation. It has achieved remarkable results on various natural image segmentation tasks and has also been applied on various medical datasets to validate their segmentation performance [21–23]. Therefore, this study adopted YOLOv8 and the SAM model to accurately localize and segment natural images of dental plaque on teeth, so that plaque scoring and indexing could be automated through deep learning.

The plaque indexing method that is commonly used in clinical practice is QHI, which are highly subjective assessments that are dependent on the expertise and calibration of the assessor. There are inconsistencies between assessors and across assessments, inaccuracies particularly for low or high plaque levels, as well as variations in the protocols used [24]. To overcome these limitations, a deep learning approach could provide dental practitioners with objective diagnoses through an efficient process.

Therefore, the overall objective of this research was to combine multiple deep learning models for the detection and segmentation of individual teeth, and the evaluation of dental health status using a CNN-based tooth plaque indexing method on intraoral images. The purpose of this study was to develop a deep learning-based indexing model Deep-Plaq, which could integrate convolutional neural network with attention-based mechanisms [25], encompassing both channel attention and spatial attention mechanisms. Spatial attention mechanisms enhanced the model's ability to understand and extract local information, which reduced the impact of background or irrelevant regions. Channel attention mechanisms assigned different weights to each channel, which helped the model focus on important channels in the feature map and thereby enhanced the expression of key features.

Materials and methods

Ethical aspects

This study received approval from the Ethics Review Board of the School of Nursing and Rehabilitation at the university (No. 2023-R-142).

Dataset

The original dataset comprised 70 subjects (20 males and 50 females) that were between 18 and 55 years old (mean age, 25 [SD, 8] years). Subjects were in good general health with the presence of anterior teeth, canines and premolars. Individuals with stained, discolored, or severely crowded dentition, large prosthodontic corrections, or those undergoing orthodontic treatment were excluded. Since anterior teeth and premolars are particularly important for their aesthetic appearance, these regions were studied to obtain valuable insights into dental plaque distribution. In contrast, molars serve a more complex role particularly in mastication, and have an indirect impact on overall oral health outcomes. Two-dimensional images from frontal and lateral views focused only on the buccal areas, gingival margins, and interproximal spaces, without accessing the lingual areas. All two-dimensional intraoral images were captured using

a Nikon D90 camera between June 1, 2021, and October 31, 2022.

The original dataset contained 210 intraoral two-dimensional images, which were comprised of frontal and lateral views of mostly permanent and a few deciduous teeth. These images were scored independently by two experts, and their high degree of agreement (Cohen's Kappa = 0.873) reflected a high level of confidence in their scoring. Teeth included were all anterior, and first and second premolars (Fig. 1a). The training, validation, and test ground-truth datasets were prepared by manually annotating teeth within images in YOLOv8 format (txt annotations and yaml config used with YOLOv8) using roboflow, an online annotation platform. Subsequently, the original dataset was processed using an algorithm, which resulted in a single-tooth dataset comprising 1400 images (Fig. 1d). Thereby, the plaque scoring challenge was converted into a six-category problem ranging from 0 to 5. The category names were utilized as folder

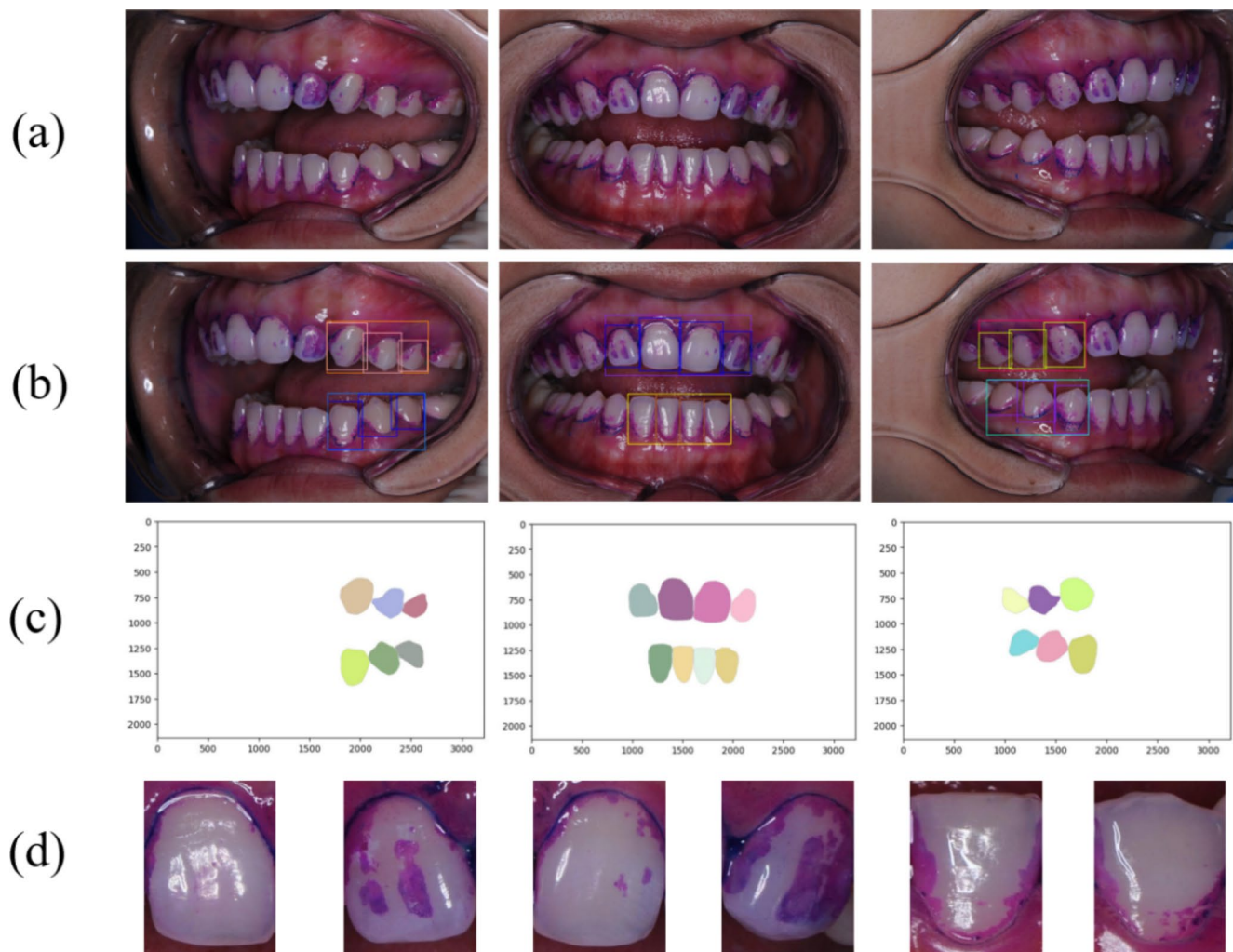
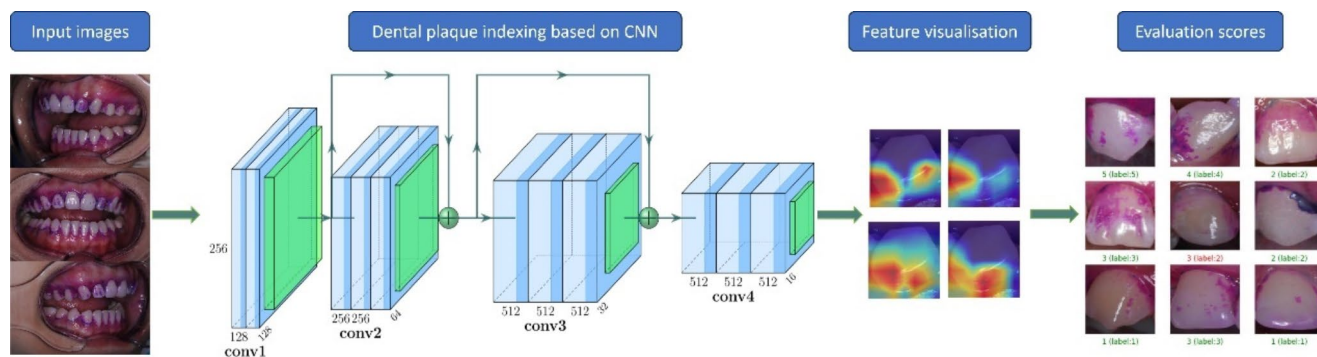
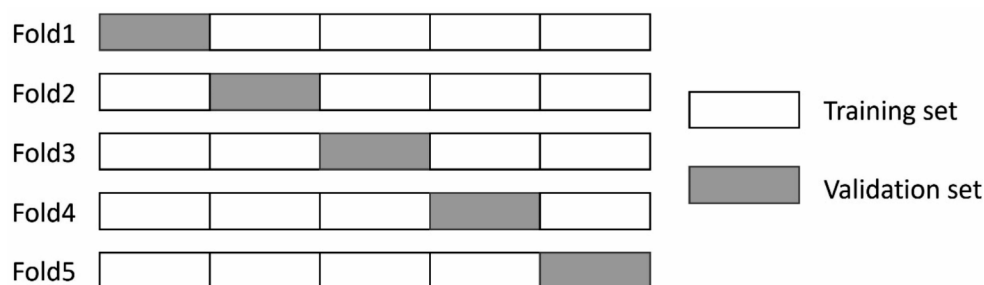


Fig. 1 Representative images from the original dataset and samples obtained following algorithmic processing. **(a)** Images from the original dataset included three different perspectives. **(b)** Images generated

by the YOLOv8 detection model. **(c)** Masks generated by the SAM segmentation model. **(d)** Images of individual teeth created by the algorithm that was implemented

Fig. 2 Schematic diagram of 5-fold cross-validation**Fig. 3** Schematic overview of the training process for dental plaque scoring of anterior and premolar dentition

names, and teeth with distinct plaque scores were organized into their corresponding folders to create a novel dataset for evaluating dental plaque scores.

During the period of tooth detection, a complete dataset was randomly divided into three sets in an 8:1:1 proportion. There were 70 photographs for each of the frontal, left lateral, and right lateral views, which were all divided according to this proportion:

- i. training set ($n = 56$), a subset for training and fitting the model with manually annotated data.
- ii. validation set ($n = 7$), a subset to evaluate performance of the model and correct the model through hyperparameter tuning.
- iii. test set ($n = 7$), a subset to test performance of the model based on training and validation data.

Thereafter, a 5-fold cross-validation method was utilized during tooth scoring, which employed one dataset for validation and the remaining four datasets for training (Fig. 2). Each fold was arbitrarily split into two subsets, as follows:

- i. training set ($n = 1120$), a subset for training and fitting the CNN model with 6 classes of data.
- ii. validation set ($n = 280$), a subset for validating performance of the model.

Region of interest detection

For an intuitive illustration of the method's workflow, a schematic diagram of the process is presented in Fig. 3. The small bounding boxes were entirely contained within the large bounding boxes for both maxillary and mandibular teeth (Fig. 1b). The large rectangular boxes provided localization by marking the entire maxillary or mandibular dentition to enhance performance of the YOLOv8 model, whereas the small rectangular boxes annotated individual teeth for detection. This model was trained on the training set with input images resized to a resolution of 640×640 pixels. Additionally, the YOLOv8 model underwent training for 300 epochs (an epoch represented a single pass through the entire training set) using four NVIDIA GeForce RTX 3090 GPUs, with an early stopping mechanism set at 50 patients to prevent overfitting when no observable improvement was detected [26]. Since YOLOv8 generated multiple predictions for single targets, non-maximum suppression (NMS) was necessary to select the most relevant bounding box among redundant overlapping boxes [27]. Intersection-over-Union (IoU) was employed as the method for filtering these bounding boxes, with the IoU value for the hyperparameter set to 0.8 and the default values of YOLOv8 maintained for the remaining hyperparameters. Since the number of multi-view intraoral images totaled only 210, a five-fold cross-validation method was employed for the identification of target teeth.

Region of interest segmentation

SAM is a prompt-based model that achieves strong zero-shot generalization, which offers practicality and flexibility for effective image segmentation across various scenarios. The powerful prompt feature of the SAM model could accurately and flawlessly obtain the mask of an individual tooth when prompted with a bounding box. Therefore, following region of interest detection, SAM was applied to segment individual teeth without the need for re-training the model. This process involved obtaining a mask from a given prompt, which was the bounding box of a single tooth (Fig. 1c). The two larger rectangular boxes served only the purpose of localization in the first stage, and did not contribute to the segmentation task of target teeth in the second stage. Consequently, these two larger boxes were filtered out through calculating the areas of rectangles. Additionally, each individual tooth was extracted using its corresponding mask, resulting in the creation of a new dataset through post-processing. Post-processing steps included assigning tooth numbers according to the FDI system. This was primarily achieved by calculating the relative position of the target tooth based on the pixel coordinates of the bounding box center points. This was intended to facilitate an ability to view and assess the quality of specific teeth. Each image contained a single tooth that was saved with a sequential name and assigned a tooth number based on the FDI system. This approach ensured the accurate and efficient identification of each tooth's associated data for subsequent utilization in assessments.

Tooth scoring

At this stage, the new dataset consisting of images of individual teeth was trained using various pre-trained CNN models. These models included the deep residual learning network (ResNet) model, along with other CNN architectures such as ResNeXt50, SqueezeNet, and VGG [28–31]. Different models were trained multiple times with different combinations of hyperparameters. Following this training with a range of CNN models, ResNeXt50 emerged as the optimal performer.

Subsequently, channel attention and spatial attention mechanisms were integrated into the ResNeXt50 architecture to create a novel DeepPlaq model, which demonstrated enhanced performance. The channel attention module aimed to amplify the feature expression within each channel, while the spatial attention module highlighted the significance of various locations within the image, thereby enhancing overall model performance. Ultimately, the DeepPlaq network outperformed other CNNs based on the same evaluation metrics.

Before training the models, data augmentation was considered an essential preparatory step [32]. All images were resized to a resolution of 256×256 pixels to address variations in fidelity, and bicubic interpolation was employed to maintain image resolution. Additionally, data augmentation techniques such as horizontal and vertical flipping, as well as random rotation (-45° to 45°), were applied during training to mitigate underfitting of the deep learning model. Regarding hyperparameters, an initial learning rate of 0.001 was utilized along with the Adam optimizer [33]. To mitigate class imbalance, the Python library *imgaug* was used to synthetically generate additional data before training, particularly to address the scarcity of samples for class 0 [34]. The cross-entropy loss function with class weights was employed, with class weights set to 0.25, 0.15, 0.1, 0.15, 0.25, and 0.1 for classes 0 to 5, respectively. The batch size was configured to 128, and the model underwent training for 30 epochs using four NVIDIA GeForce RTX 3090 GPUs. A comprehensive workflow of this study is shown in Fig. 4. Additionally, a logistic regression model was implemented by converting the image into the HSV color space. Mean Absolute Error (MAE) was calculated as the evaluation metric and compared with the DeepPlaq model.

Classification performance and model visualization

During the evaluation phase of the plaque scoring network, each model underwent comprehensive assessment using four common classification evaluation metrics: precision, recall, accuracy, and F1 score. Additionally, these evaluation metrics were calculated using a weighted average to address the issue of class imbalance in multi-classification problems. Based on the aforementioned metrics, the performance of the CNNs was evaluated according to specific performance measures outlined in Table 1.

The explanatory visualization technique known as gradient-weighted class activation mapping (Grad-CAM) was introduced to visualize features. Grad-CAM utilizes gradients derived from a target concept to generate a rough localization map, highlighting crucial regions in the image for concept prediction. In the Grad-CAM heat map, regions appearing in a deeper red indicated a higher reliance of the model on features within that area for discrimination. This process relies on the fundamental principle of Grad-CAM, where it computes the importance weight of each feature map in the last convolutional layer with respect to the image class. It then aggregates these weighted sums across all feature maps, and maps them back onto the original image.

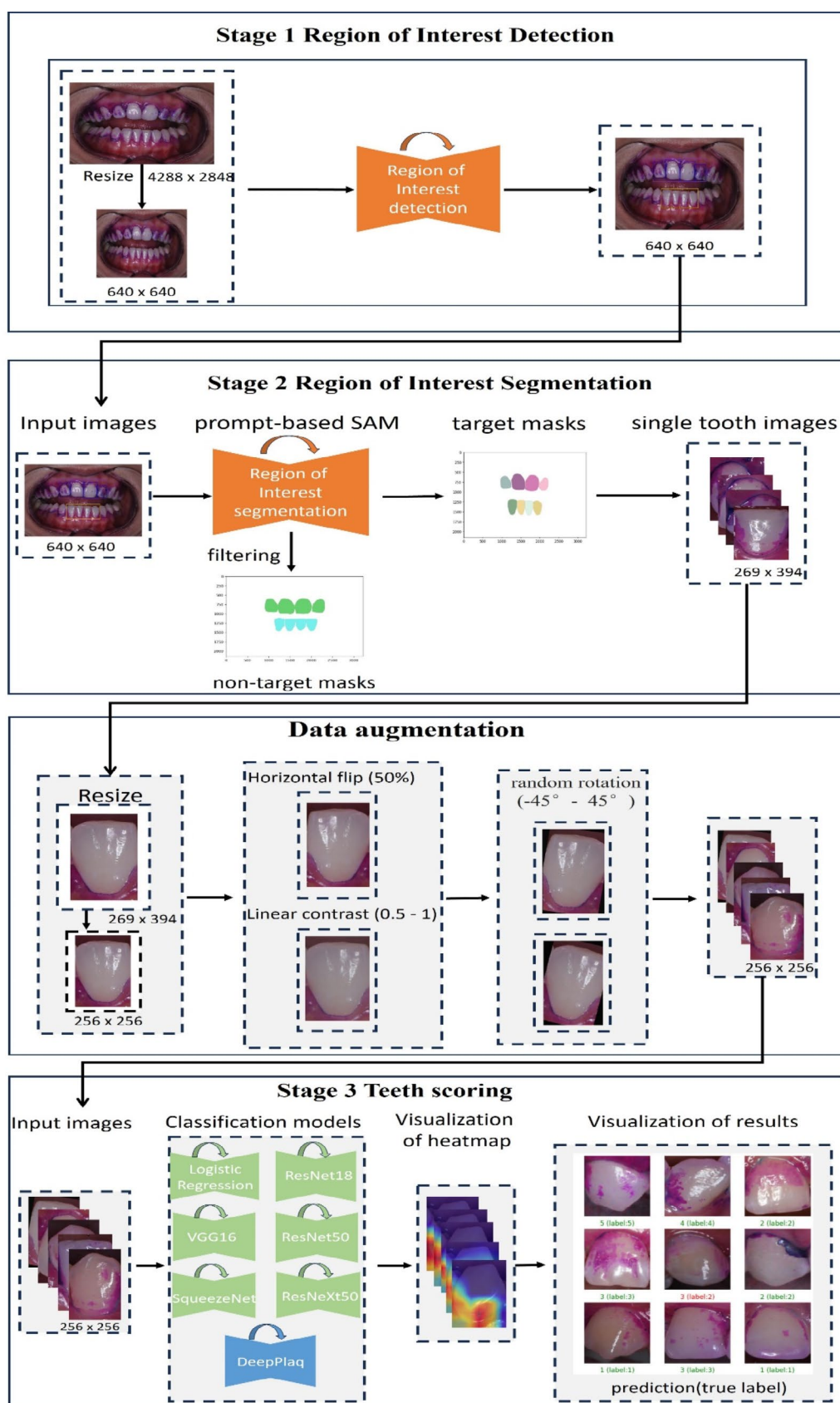


Fig. 4 Pipeline of the region of interest detection, region of interest segmentation, data augmentation and teeth scoring procedures

Table 1 The performance measures of CNN for classification evaluation

Metrics	Formula	Evaluation Focus
Accuracy (acc)	$\frac{TP+TN}{TP+TN+FP+FN}$	Ratio of correct predictions over total number of instances evaluated.
Precision (P)	$\frac{TP}{TP+FP}$	Positive patterns that are correctly predicted from total predicted patterns in a positive class.
Recall (R)	$\frac{TP}{TP+FN}$	Fraction of positive patterns that are correctly classified.
F1 Score (F1)	$\frac{2 \times P \times R}{P+R}$	Combining precision and recall to assess a classification model's performance.
Weighted Precision	$\sum_{i=1}^l P_i \frac{n_i}{N}$	The weight of per-class precision
Weighted Recall	$\sum_{i=1}^l R_i \frac{n_i}{N}$	The weight of per-class recall
Weighted F1 Score	$\sum_{i=1}^l F1_i \frac{n_i}{N}$	The weight of per-class F1 score

Note - TP_i: true positive for class i; TN: true negative; FP: false positive; FN: false negative; n_i: number of samples of for class i; N: total number of samples $\hat{y}_i$: model prediction y_i : true label

Table 2 Precision, recall, accuracy, and F1-score of CNNs evaluating the plaque levels on teeth

Network model	Precision	Recall	Accuracy	F1-Score
LogisticRegression	0.451	0.455	0.455	0.451
VGG-16	0.506	0.532	0.532	0.502
SqueezeNet	0.706	0.592	0.592	0.618
ResNet18	0.742	0.740	0.740	0.736
ResNet50	0.756	0.746	0.746	0.74
ResNeXt50	0.766	0.764	0.764	0.762
DeepPlaq	0.784	0.776	0.776	0.774
(ResNext50 + Channel-Spatial Attention)				

Results

Detection and segmentation performance

The mAP of YOLOv8 in detecting regions of interest was approximately 0.941 ± 0.005 . Specifically, the mAP for anterior teeth was up to 0.947, whereas for the left and right teeth it was 0.946 and 0.936, respectively. The segmentation task stage took advantage of characteristics of SAM that can produce high-quality object masks when prompted with a box, so that the segmentation performance depended entirely on the detection performance of the previous stage.

Classification performance and model visualization

The performance of multiple deep learning models on two-dimensional images that each contained a single tooth were evaluated. A 5-fold cross-validation method for each CNN model was used to calculate the best precision, recall, accuracy, and F1-score of evaluating plaque levels on teeth, as depicted in Table 2. Additionally, these metrics were calculated for a linear classification model called Logistic Regression, demonstrating superior performance of the CNN models. Multiple training iterations were conducted for each CNN model due to varying hyperparameter

combinations, and the average results for each metric were recorded using the 5-fold cross-validation method. The confusion matrix, depicted in Fig. 5, provided a more intuitive representation of the classification performance of each model. Furthermore, the presented confusion matrix was a cumulative result obtained by summing the matrices from each fold, given the utilization of a 5-fold cross-validation method during training. The DeepPlaq model was compared with a logistic regression model on each fold. The comparison results between the top-performing DeepPlaq model and the logistic regression model are shown in Fig. 6.

Clinical implications

Deep learning models, after extensive training on data, can accurately detect the distribution and quantity of dental plaque on teeth, thereby reducing bias and errors caused by subjective factors. Expedient and accurate detection of plaque could lead to earlier identification of dental caries, gingivitis, and periodontal disease, and thereby the implementation of appropriate intervention strategies to prevent and manage the progression of disease. Readily accessible visualization of a patient's plaque can enhance patient's compliance and motivation in the prevention and management of dental disease. Additionally, the collection and analysis of large amounts of dental plaque data through deep learning could reveal new clinical patterns, provide valuable data for research, and further optimize clinical decision-making.

These pre-trained models were used with the 5-fold cross-validation method for training, resulting in 5 optimal checkpoint files of models. Subsequently, these 5 trained checkpoints were utilized to compute the feature map for an image, which allowed visualization of the model's capability in feature localization. In the activation map, the DeepPlaq model demonstrated notable proficiency in focusing on plaque characteristics on both sides of the teeth in each fold.

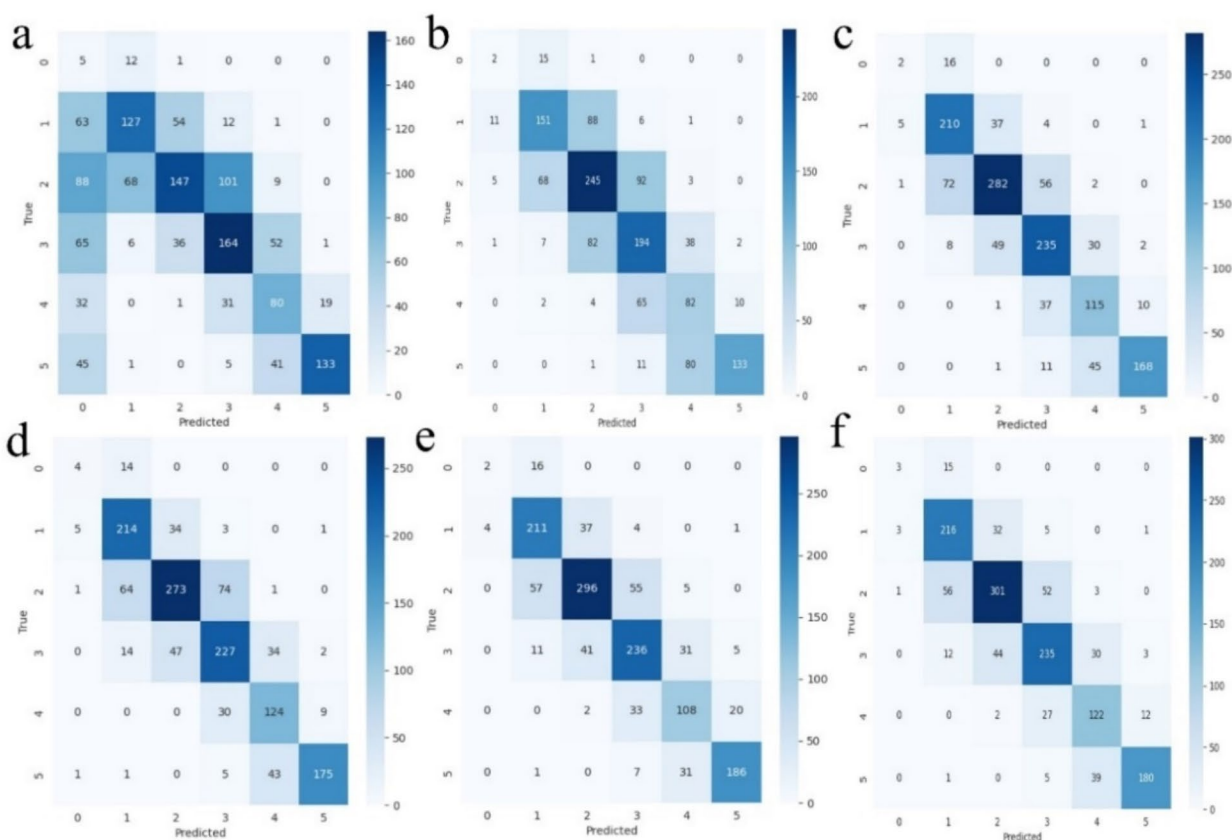
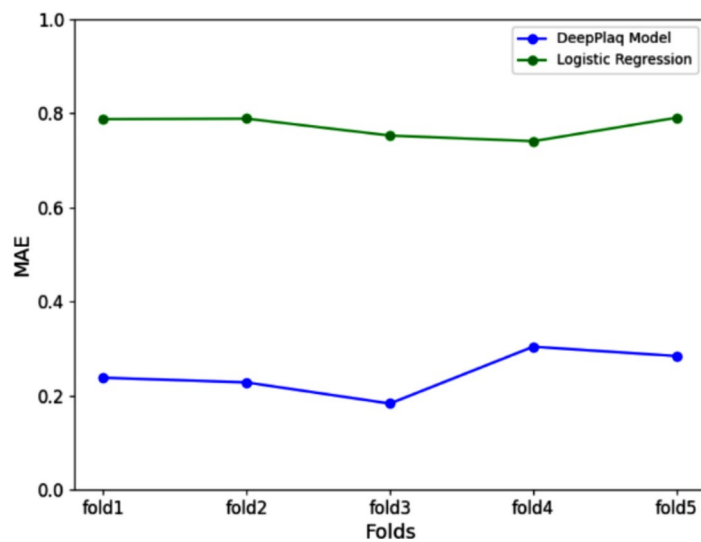


Fig. 5 The confusion matrices of CNNs classifying classes of teeth with plaque display agents: (a) VGG16, (b) SqueezeNet, (c) ResNet18, (d) ResNet50, (e) ResNeXt50, (f) DeepPlaq

Fig. 6 The comparative result of MAE between DeepPlaq and Logistic Regression model



Conversely, other CNNs did not exhibit clear patterns in the highlighted areas (Fig. 7).

Discussion

Principal results

This study demonstrated that multi-view intraoral images combined with deep learning technology through pre-trained DL models could effectively classify dental plaque levels, and achieve an accuracy of up to 0.776. The study utilized a dataset of two-dimensional intraoral images to evaluate plaque distribution and severity on anterior and premolar teeth. The YOLOv8 model was used to detect regions of interest, and then the SAM model utilized bounding boxes as prompts to segment the target teeth. This approach fully leveraged the prompting capabilities of the SAM model, thereby achieving excellent performance in both detection and segmentation tasks. The application of CNNs to evaluate dental plaque distribution could ultimately expedite diagnostic processes and enhance the efficiency of patient management. Although this study was restricted to intraoral images of individual anterior and premolar teeth, this application could be extended to the remaining dentition, other assessments of hard and soft tissues in the oral cavity, as well as various different imaging modalities. These

developments will ultimately support dentists with their diagnostic tests, treatment planning processes, and optimal delivery of treatment protocols.

Limitations

The current study had some limitations that need to be addressed in future work. First, the dataset for this study included only healthy individuals aged 18 to 55 who had a full set of anterior teeth, first premolars, and second premolars. However, these subjects lacked diversity in their socioeconomic, occupational and racial backgrounds. Future studies should endeavor to collect a larger and more diverse dataset to improve performance of models, which would enhance the generalizability and versatility of this classification model. Second, this study did not assess plaque on the lingual surfaces of teeth. In future research, appropriate methods for collecting data from the lingual areas will need to be evaluated. Third, the theoretical research conducted in this study was fully automated, but is yet to be developed into a practical system. Therefore, this work will be developed into a practical application for clinical use in an ongoing effort to support dentists with diagnoses, treatment planning, and patient management. Finally, this classification model still requires further improvements to enhance its performance and achieve superior results.

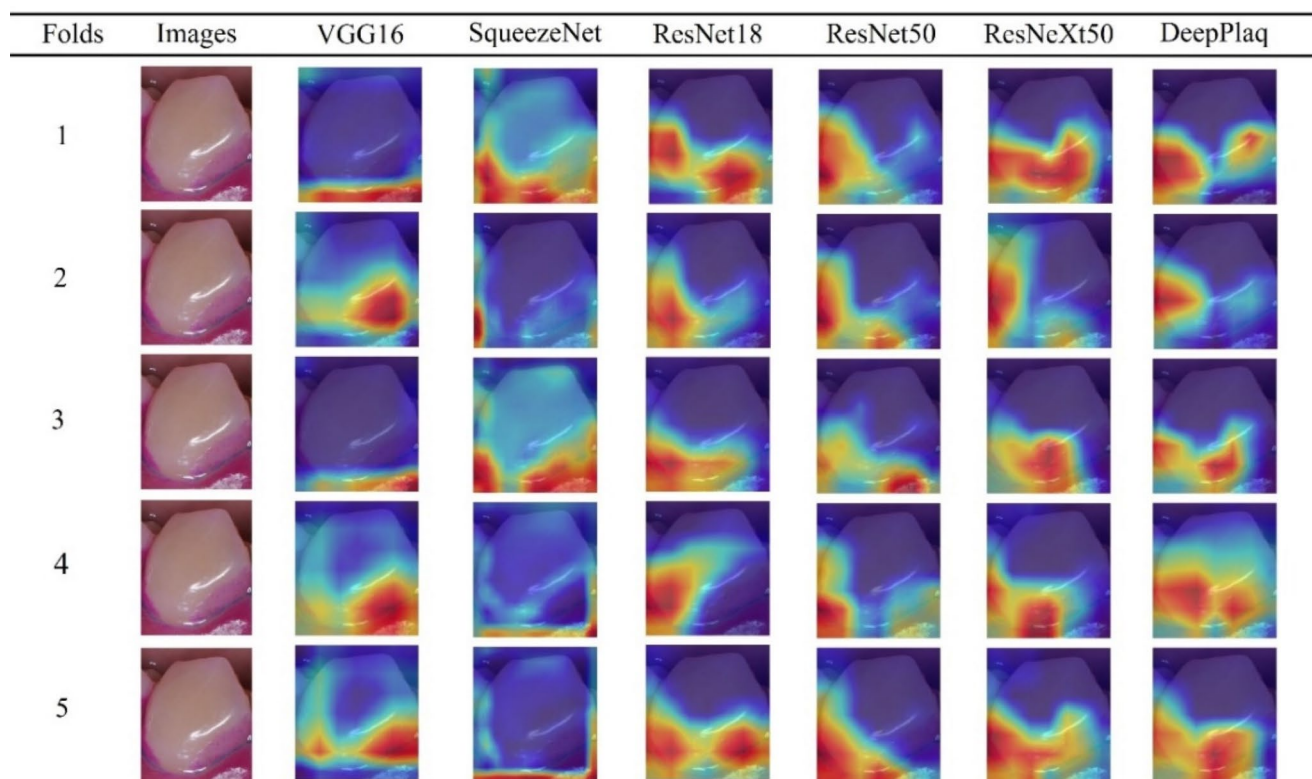


Fig. 7 Feature visualization of five-fold cross validation for each model

Comparison with prior work

Evaluations of dental plaque distribution and severity on teeth with plaque disclosing agents are usually assessed subjectively by experienced experts. However, dental plaque can now be scored and indexed objectively through the use of deep learning methods for detection, segmentation, and classification. Previous studies utilized a variety of image processing techniques for the detection and segmentation of dental plaque. For instance, You et al. [35] used a visual object classes dataset for pretraining the basic DeepLab network, and then the DeepLabV3+ model was trained using an intraoral image dataset of primary teeth to detect dental plaque. However, this method was not applied to permanent dentition. Liu et al. [36] used Keras, a deep learning framework, to build DeeplabV3+ for a multi-class semantic segmentation model to extract tooth contours from intraoral images, and then identify dental plaque based on HSV value. But this approach was designed to compute the plaque index for multiple teeth within an image, and lacked implementation for assessing plaque from images of individual teeth. Li et al. [37] proposed a novel low-shot learning method for intelligent dental plaque segmentation directly from oral endoscopic images without staining reagents. The proposed method achieved an IOU of 85.9% for plaque and the corresponding teeth, whereas experienced dentists' evaluation yielded an IOU of 69.4%. Indeed, this method exhibited outstanding performance on small-scale datasets. Then Li et al. [38] proposed a new network enhanced by a self-attention module and local-to-global features fusion for intelligent dental plaque segmentation. Their study aimed to utilize intraoral images bypassing the need for dyeing reagents, and the method achieved outstanding performance in the tooth segmentation task. Andrade et al. [39] employed U-Net network to segment the intra-oral images after applying dental biofilm-disclosing agents and achieved a good result. But unlike these previous approaches, the current study evaluated dental plaque on teeth by using multi-view intraoral images. This approach first used YOLOv8 for detecting target teeth, and then fully leveraged the prompt capabilities of SAM to segment target teeth, thereby obtaining a dataset of individual teeth. Subsequently, the quality of each tooth could be assessed by using deep learning classification methods, which enables the assessment of specific teeth by their dentist.

Conclusions

This study introduced a deep learning-based automated method for detecting, segmenting, numbering, and evaluating dental plaque distribution on teeth in two-dimensional

intraoral images. This algorithm comprised a YOLOv8 model for tooth detection, a SAM-based model for tooth segmentation, a post-processing algorithm for tooth numbering, and a CNN-based classification model named DeepPlaq for evaluating plaque distribution on teeth. There was exceptional performance in accurately detecting, segmenting, and numbering teeth. Moreover, the evaluation of dental plaque index using the CNN-based classification model yielded promising results. This study marked the first attempts to detect and segment teeth in the original dataset, resulting in a dataset of single-tooth images. Subsequently, the classification model was utilized to categorize plaque levels based on QHI. This plaque index algorithm holds promise in advancing automated programs for dental plaque assessment, which could expedite diagnosis and treatment planning in oral health care.

Author contributions Xu Chen: Methodology, Software, Investigation, Formal analysis, Visualization, Data curation, Writing - original draft, Writing - review & editing Yiran Shen: Conceptualization, Project administration, Funding acquisition, Methodology, Supervision, Validation, Writing - review & editing Jin-Sun Jeong: Methodology, Software, Investigation, Formal analysis, Visualization, Data curation, Writing - review & editing Hiran Perinpanayagam: Supervision, Validation, Writing - review & editing Kee-Yeon Kum: Supervision, Validation, Writing - review & editing Yu Gu: Conceptualization, Resources, Supervision, Validation, Writing - review & editing.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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